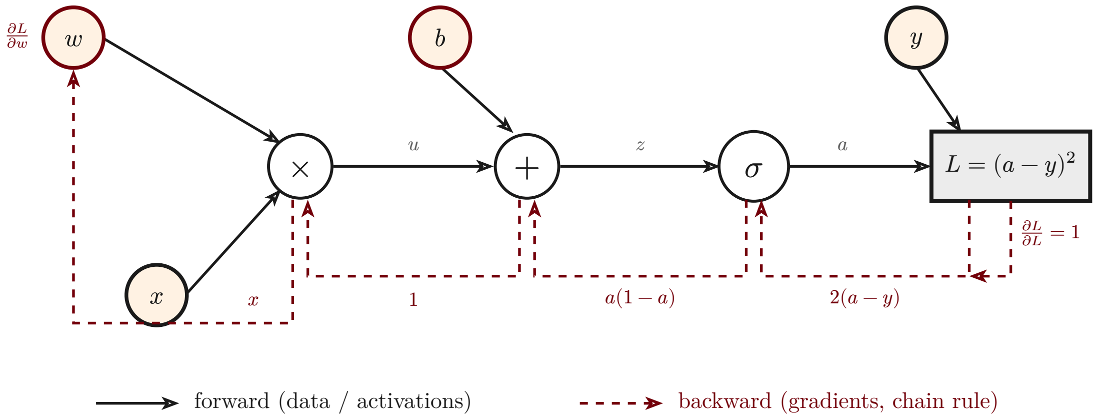
Computational graph: forward pass and backpropagation

$$u = w \cdot x, \quad z = u + b, \quad a = \sigma(z), \quad L = (a - y)^2$$



Each op caches its inputs on the forward pass. Backprop seeds $\partial L/\partial L = 1$ at the loss and walks the DAG in reverse: every dashed edge multiplies by the **local** derivative of its op, so the global gradient at each parameter (e.g. $\partial L/\partial w$) is the product of the local factors along its path — the chain rule.